CODEALPHA

DATA ANALYTICS INTERNSHIP

ONLINE RETAIL SALES ANALYSIS

Exploratory Data Analysis (EDA) & Data Visualization Using Python

COMPLETED TASKS

Task 2 — Exploratory Data Analysis (EDA)
Task 3 — Data Visualization

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Branch	B.Tech — Artificial Intelligence & Data Science (AIDS)
University	Savitribai Phule Pune University (SPPU)
Internship Domain	Data Analytics
Company	CodeAlpha
Internship Period	8 September 2026 – 26 September 2026
Team	Individual Project
GitHub Repository	github.com/hemangikalyanshetti/CodeAlpha_Data_Analytics

Dataset: UCI Online Retail Dataset
Tools: Python • Pandas • Matplotlib • Seaborn

Prepared as an internship project report

1. Project Overview

This project analyzes an online retail transaction dataset to identify sales patterns, top-performing products and countries, monthly revenue trends, and potential data-quality issues. The analysis demonstrates practical skills in data loading, cleaning, exploratory analysis, aggregation, visualization, and interpretation using Python.

2. Objectives

• Understand the structure and quality of the retail dataset.
• Clean records unsuitable for sales calculations.
• Calculate revenue, orders, units sold, and average order value.
• Identify top products and countries by revenue.
• Examine monthly revenue patterns.
• Present findings through clear visualizations and business-oriented insights.

3. Dataset Description

The workbook contains 541,909 transaction rows. The main fields include InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. 135,080 rows have a missing CustomerID. For sales analysis, records with non-positive quantity or unit price were excluded.

4. Technology Stack

Python, Pandas, Matplotlib, and Seaborn were used for data manipulation, statistical exploration, aggregation, and visualization.

5. Exploratory Data Analysis

The EDA stage focused on data structure, missing values, validity of numeric fields, and summary metrics. Revenue was calculated as Quantity × UnitPrice. For business sales analysis, only positive quantity and positive unit-price records were retained.

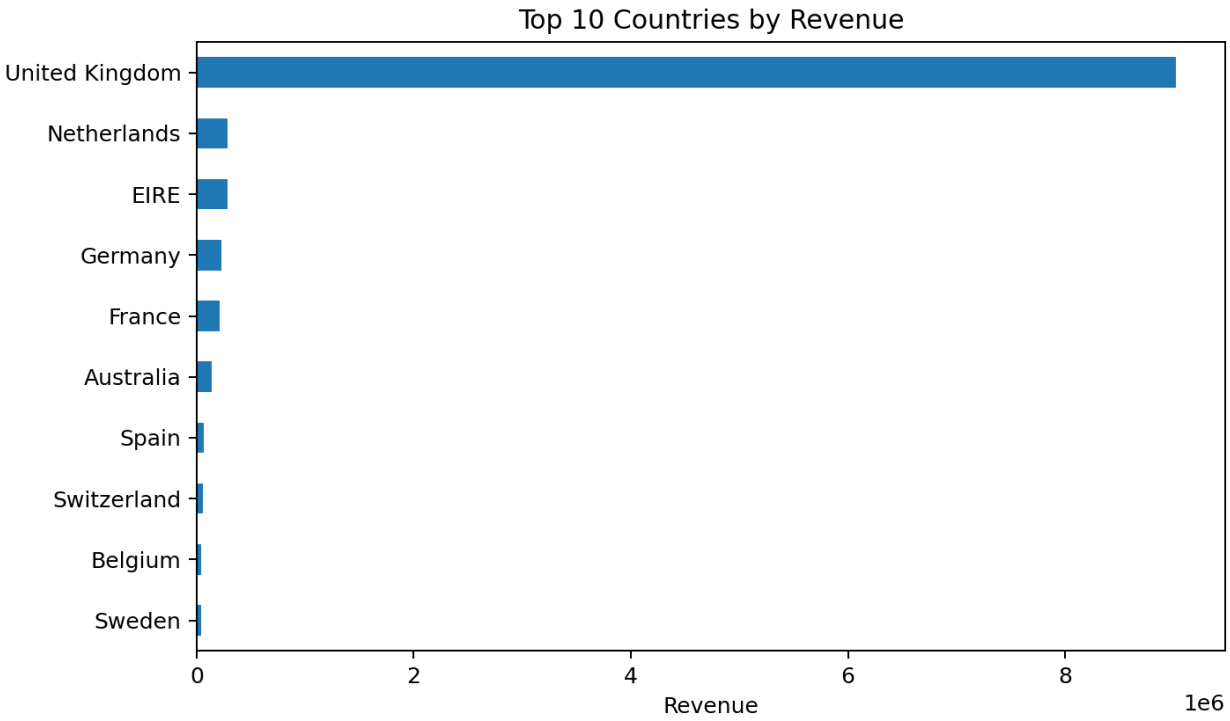
Metric	Result
Raw transaction rows	541,909
Valid positive-sales rows	530,104
Total revenue	£10,666,684.54
Total units sold	5,588,376
Unique orders	19,960
Unique customers	4,338
Average order value	£534.40

Key EDA observations

• The dataset contains 541,909 raw rows and 530,104 rows after positive-sales filtering.
• Total analyzed revenue is £10,666,684.54.
• There are 19,960 unique invoices/orders and 4,338 unique customers.
• Missing CustomerID values are present and should be considered for customer-level analysis.
• Transaction-level revenue supports comparisons across products, countries, and time periods.

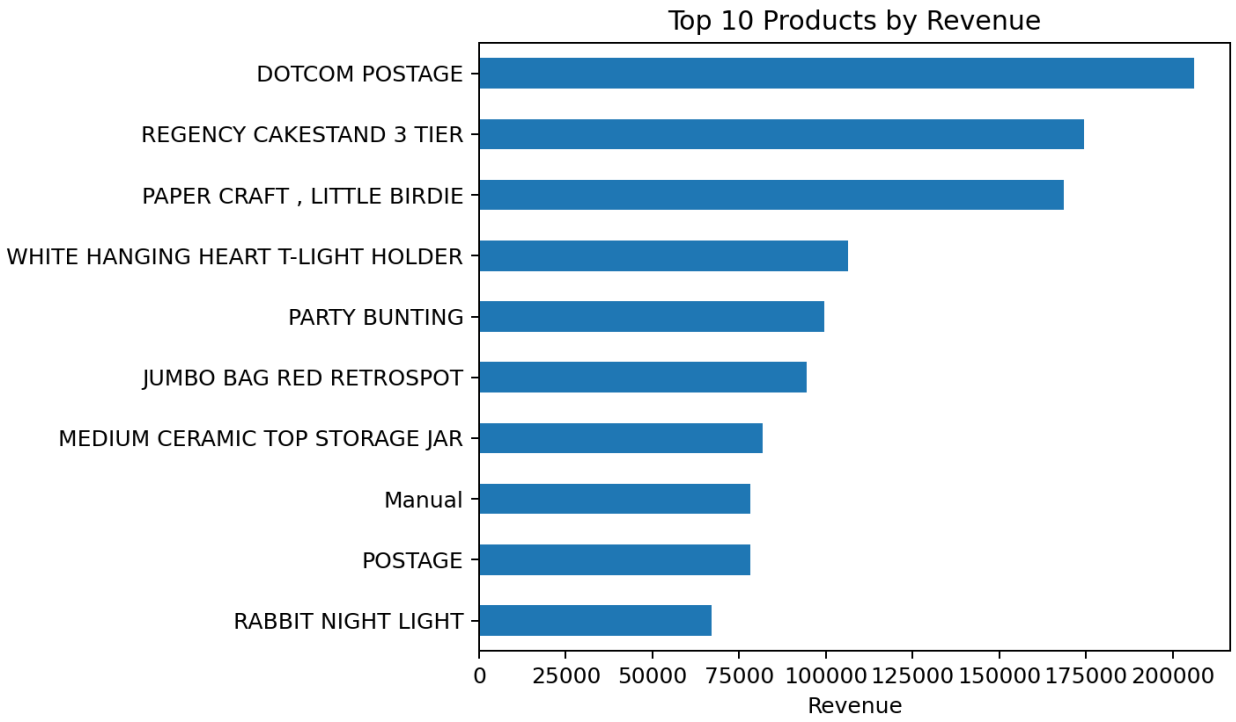
6. Data Visualization

Top 10 Countries by Revenue



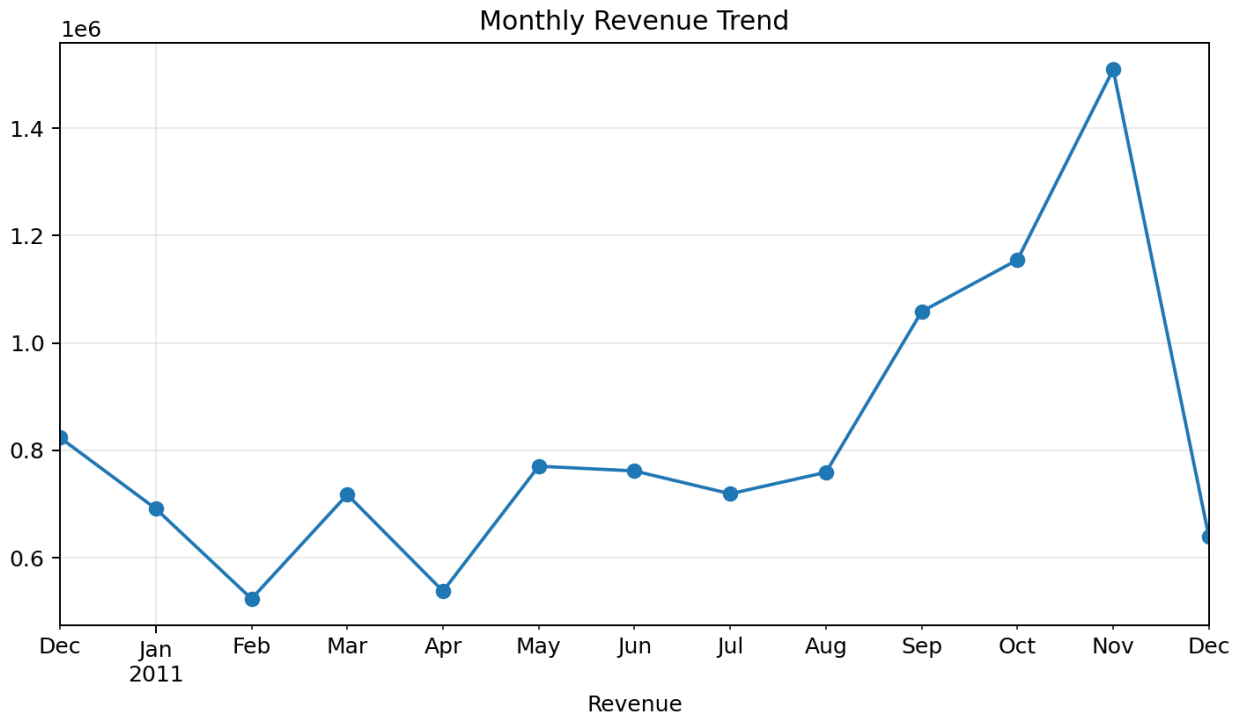
The leading country by revenue is United Kingdom, with revenue of £9,025,222.08.

Top 10 Products by Revenue



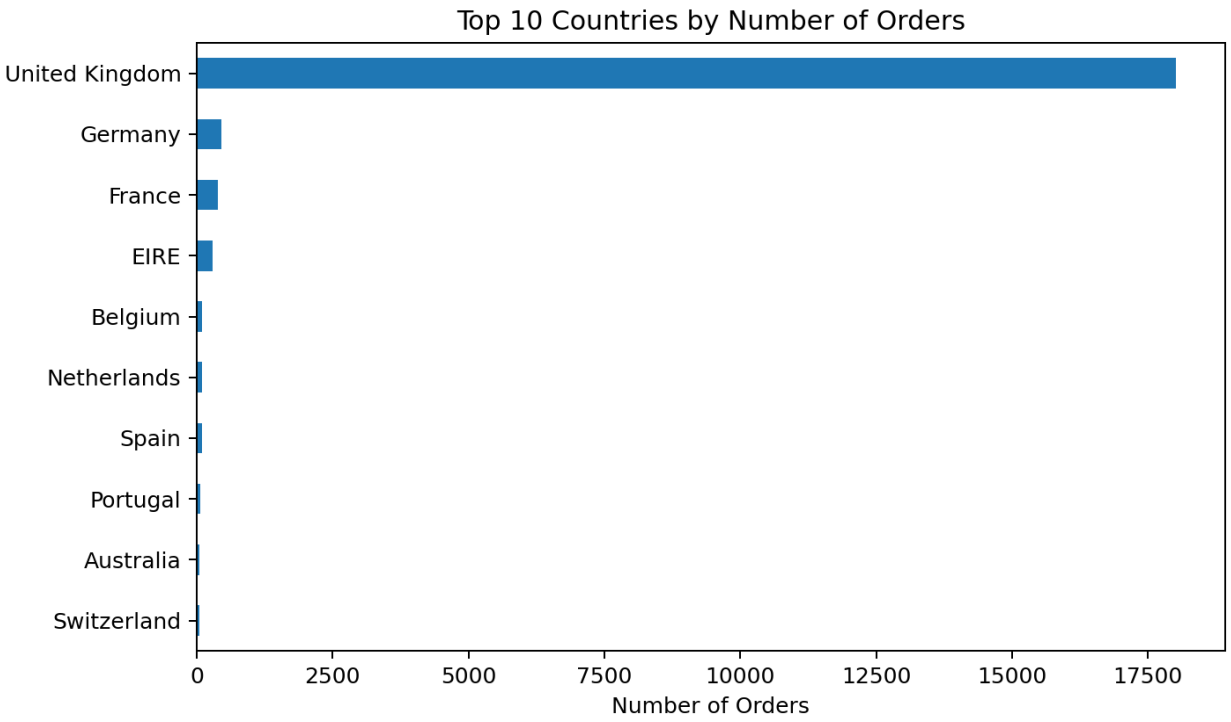
The highest-revenue product is “DOTCOM POSTAGE”, generating £206,248.77.

Monthly Revenue Trend



Monthly aggregation provides a time-based view of sales activity and can support seasonal or promotional analysis.

Orders by Country



United Kingdom has the largest number of orders among the countries shown.

7. Business Insights

1. Revenue is not evenly distributed across countries, allowing market-level performance to be compared.
2. A small group of products contributes strongly to product revenue, making product-level monitoring useful.
3. Monthly revenue trends provide a foundation for seasonal and campaign analysis.
4. Missing customer identifiers limit some customer-level analyses.
5. Comparing order count with revenue gives a more complete view of market performance.

8. Conclusion

The project demonstrates a practical introductory data analytics workflow: understanding a real dataset, cleaning records, calculating business metrics, exploring patterns, and communicating findings through visualizations. The analysis can be extended with customer segmentation, cohort analysis, forecasting, or an interactive dashboard.

9. CodeAlpha Task Mapping

Task 2 — Exploratory Data Analysis: Dataset structure, data types, missing values, summary statistics, trends, and data-quality issues were examined.

Task 3 — Data Visualization: Revenue, product, country, order-volume, and monthly trend data were transformed into charts to communicate findings clearly.

10. Reproducibility

The report was generated directly from the uploaded Online Retail.xlsx workbook. The analysis uses the rule Quantity > 0 and UnitPrice > 0 for positive sales and computes Revenue = Quantity × UnitPrice.

Before submission: Add your name, college, internship dates, and GitHub repository link on the cover page if required by your coordinator.